A Hamiltonian-Path Criterion from Bipartite Number and Average Eccentricity

A proof of Written on the Wall II, Conjecture 198a

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Abstract. For a finite nontrivial connected simple graph G, let $b(G)$ be the maximum order of an induced bipartite subgraph and let $\overline{\mathrm{ecc}}(G)$ be the average vertex eccentricity. We prove that

$$b(G) \le 2 + \overline{\mathrm{ecc}}(G) \quad \Longrightarrow \quad G \text{ has a Hamiltonian path.}$$

This is the statement recorded as Written on the Wall II, Conjecture 198a. The proof is driven by the tight chain $D+1 \le b(G) \le 2+\overline{\mathrm{ecc}}(G) \le D+2$, where $D = \mathrm{diam}(G)$. In the first equality case, every vertex outside a diametral path belongs to a clique layer that can be inserted into that path. In the second, the graph is self-centered of diameter two, and the Chvátal–Erdős traceability theorem applies.

1. Statement and notation

All graphs in this note are finite and simple. For a connected graph G, write

$$D = \mathrm{diam}(G), \qquad r = \mathrm{rad}(G), \qquad \overline{\mathrm{ecc}}(G) = \frac{1}{|V(G)|} \sum_{v \in V(G)} \mathrm{ecc}_G(v).$$

The *bipartite number* $b(G)$ is the largest cardinality of a set $S \subseteq V(G)$ for which the induced graph $G[S]$ is bipartite. A graph is *traceable* if it has a Hamiltonian path.

Theorem 1 (Written on the Wall II, Conjecture 198a)**.**
Let G be a nontrivial connected graph. If

$$b(G) \le 2 + \overline{\mathrm{ecc}}(G),$$

then G is traceable.

We use two classical facts. First,

$$b(G) \ge 2r, \tag{1}$$

as recorded by DeLaViña, Pepper, and Waller [DPW07, Theorem 2]. Second, the Chvátal–Erdős path theorem says that an s-connected graph with no independent set of $s+2$ vertices is traceable [CE72, Theorem 2].

2. Proof

Choose a diametral geodesic

$$P = v_0 v_1 \cdots v_D.$$

A geodesic is induced, so P is an induced bipartite graph of order $D+1$. Since every eccentricity is at most D,

$$D+1 \le b(G) \le 2 + \overline{\mathrm{ecc}}(G) \le D+2. \tag{2}$$

Because $b(G)$ is integral, only two cases remain.

Case 1: $b(G) = D+1$

Fix $x \notin V(P)$. The induced graph on $V(P) \cup \{x\}$ has order $D+2 > b(G)$, hence is not bipartite. Thus x has neighbors on P of both parities. On the other hand, any two P-neighbors v_i, v_k of x satisfy $|i-k| \le 2$: otherwise $v_i x v_k$ would shorten the corresponding section of the geodesic P.

Let j be the least index for which $xv_j \in E(G)$. The preceding paragraph gives

$$N_G(x) \cap V(P) \in \big\{\{v_j, v_{j+1}\}, \{v_j, v_{j+1}, v_{j+2}\}\big\}, \tag{3}$$

where the second alternative is omitted when $j+2 > D$. For $0 \le j < D$, let X_j consist of the vertices outside P whose least neighbor index is j.

Each X_j is a clique. Indeed, if distinct $x, y \in X_j$ were nonadjacent, then

$$\big(V(P) \setminus \{v_{j+1}\}\big) \cup \{x, y\}$$

would induce a bipartite graph of order $D+2$. To see the bipartition directly, retain the alternating coloring of P: the only remaining possible P-neighbors of x or y are v_j and v_{j+2}, which have the same color, and assign both x and y the opposite color. This contradicts $b(G) = D+1$.

Now order the vertices as

$$v_0,\ X_0,\ v_1,\ X_1,\ \ldots,\ v_{D-1},\ X_{D-1},\ v_D. \tag{4}$$

using any order inside each X_j and simply omitting empty layers. Consecutive vertices are adjacent: every X_j is a clique, and every member of X_j is adjacent to both v_j and v_{j+1}. All vertices occur exactly once, so (4) is a Hamiltonian path.

Case 2: $b(G) = D+2$

Equality throughout the right half of (2) gives $\overline{\mathrm{ecc}}(G) = D$. Every vertex therefore has eccentricity D, so G is self-centered and $r = D$. Combining this with (1) yields

$$2D = 2r \le b(G) = D+2,$$

and hence $D \le 2$. The case $D = 1$ is impossible: a nontrivial diameter-one graph is complete and has bipartite number 2, not $D+2 = 3$. Consequently,

$$D = r = 2, \qquad b(G) = 4. \tag{5}$$

The graph G has no cut vertex. Otherwise, let c be a cut vertex. For any $w \ne c$, choose y in a component of $G - c$ different from the component containing w. Every w-y path passes through c, and a path of length at most two must therefore be wcy. Hence every vertex other than c is adjacent to c, giving $\mathrm{ecc}_G(c) = 1$, contrary to self-centeredness with radius two. Since $D = 2$ also implies $|V(G)| \ge 3$, the graph is 2-connected.

Finally, if I is a maximum independent set, then $I \ne V(G)$ because G is connected and nontrivial. Adding any vertex outside I produces an induced bipartite graph, so $b(G) \ge |I| + 1$. By (5), $\alpha(G) \le 3$. Hence this 2-connected graph has no independent set of four vertices. The Chvátal–Erdős path theorem, with $s = 2$, supplies a Hamiltonian path. This completes the proof of Theorem 1. $\square$

3. Formalization and reproducibility

The theorem is represented in the Formal Conjectures repository over a finite type α by a connected simple graph G, with $b(G)$ coerced to $\mathbb{R}$ and average eccentricity defined as a real average. The proof artifact accompanying this manuscript is organized around the following independently checkable interfaces:

1. a diametral geodesic and the lower bound $D+1 \leq b(G)$;
2. the integral dichotomy $b(G) \in \{D+1, D+2\}$;
3. the attachment and clique-layer lemmas (3);
4. the self-centered diameter-two reduction; and
5. a direct longest-path traceability endpoint matching the repository statement:

$$\exists a\, b, \quad \exists p : G.\mathrm{Walk}\ a\ b, \quad p.\mathrm{IsHamiltonian}.$$

All of these components, including both equality branches and the classical inequality (1), are checked in Lean without proof placeholders. Their top-level composition has the exact hypotheses and conclusion of the repository conjecture.

Artifact status. Lean 4.27 kernel checks pass for the exact repository theorem, including $b(G) \geq 2\,\mathrm{rad}(G)$, with no proof placeholders.

Research process and use of generative AI. I used an autonomous generative-AI research workflow to identify and investigate Conjecture 198a. The workflow began with a comparative analysis of problems reported as solved during the preceding several days, their statements, and the proof strategies used. This analysis was used to assess related open conjectures, after which I selected Conjecture 198a for sustained investigation.

OpenAI GPT-5.6 Sol was used through local Codex at extra-high and ultra reasoning settings for exploratory research, proof development, Lean formalization, verification, and repository preparation. Cloud-hosted ChatGPT workflows and individual ChatGPT 5.6 Pro sessions were also used for research, proof development, and verification. Claude Fable 5 and Claude Sonnet 5 were used for limited supplementary searches. These systems are disclosed as research tools and are not authors. I take responsibility for the manuscript and submitted material.

Prior-solution search and research status

The statement appears as Conjecture 198a in Written on the Wall II [DeL] and in the Formal Conjectures snapshot [The26]. A targeted search completed on 23 July 2026 covered the current upstream statement, its open and closed issue and pull-request history, public GitHub code indexed under the exact theorem identifier, and exact-phrase web queries. It found copies of the open conjecture and benchmark tasks, but no evidence in the searched sources of a proof or verified formalization predating this work. The search was not an exhaustive literature review, does not establish worldwide novelty or priority, and supports no claim of a first proof.

References

[CE72] Václav Chvátal and Paul Erdős. A note on hamiltonian circuits. *Discrete Mathematics*, 2(2):111–113, 1972.

[DeL] Ermelinda DeLaViña. Written on the wall ii: Conjectures of graffiti.pc. Online conjecture list. Conjecture 198a.

[DPW07] Ermelinda DeLaViña, Ryan Pepper, and Bill Waller. Independence, radius and hamiltonian paths. *MATCH Communications in Mathematical and in Computer Chemistry*, 58(2):481–510, 2007.

[The26] The Formal Conjectures Authors. Written on the wall ii, graph conjecture 198a. Formal Conjectures repository, 2026. Snapshot commit e751934294a381afd2d5fc1124c5953c8e25f9fa, accessed 23 July 2026.